Task-Oriented Multimodal Eye-Gaze and Voice Control of a 5-DoF Assistive Robotic Manipulator

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***Abstract*—** This paper presents a task-oriented multimodal control strategy for a five-degree-of-freedom assistive robotic manipulator equipped with an actuated gripper. The proposed approach combines eye-gaze tracking and speech recognition, assigning eye gaze to directional control and voice commands to gripper actuation and the activation of automatic routines. Eye-gaze tracking was implemented in Python using MediaPipe Face Mesh, while MATLAB performed the recognition of the commands ABRIR, CERRAR, UNO, DOS, and TRES, as well as kinematic coordination and trajectory generation. Joint positions were transmitted to an ESP32 to drive the servomotors through PWM signals. Experimental evaluation showed an overall accuracy of 96.4% for eye-gaze classification and 98.0% for speech recognition, with a total latency of 102 ms. The manipulator achieved a mean Cartesian error of 2.6 mm, a success rate of 98.3% in the automatic routines, an average execution time of 9.4 s, and a repeatability RMSE of 2.3 mm. Furthermore, a survey conducted with twenty participants yielded an overall mean score of 3.90/5, with 81.5% of the ratings corresponding to categories 4 and 5. The results support the technical feasibility of the proposed multimodal strategy for basic manipulation tasks under controlled conditions.

Keywords*—assistive robotics, human–robot interaction, eye-gaze tracking, speech recognition, multimodal control, robotic manipulator.*

I. INTRODUCTION

Object manipulation is a fundamental capability for performing everyday activities such as reaching, grasping, transporting, and placing objects in the environment. In individuals with upper-limb motor impairments, the ability to perform these actions may be considerably reduced, increasing their dependence on others and limiting their ability to perform daily activities [1]. For this reason, assistive robotics has promoted the development of robotic manipulators capable of assisting users in performing a variety of physical tasks. However, their effectiveness depends not only on the mechanical structure of the system or the accuracy of its movements, but also on the interface used to interpret the user's intentions [2]. Conventional input devices, such as joysticks, keyboards, or touch panels, require coordination and voluntary hand movements and are therefore not well suited for users with severe motor impairments [3].

Eye gaze represents another non-manual interaction modality, as it enables the interpretation of where the user is looking without requiring the operation of physical control devices [4]. In robotics, eye gaze can be used to select targets, issue motion commands, or provide information about the user's intended movement [5]. However, the performance of these systems depends on factors such as calibration, head posture, illumination, image quality, and accurate gaze-point detection. If eye gaze is used simultaneously to observe the environment, control the robot, and confirm actions, the user's visual workload may increase. Furthermore, unintended activations may occur when a natural gaze fixation is incorrectly interpreted as a command [6].

Voice provides an additional interaction modality for issuing commands without using the upper limbs or diverting attention from the workspace [7]. Voice commands can be associated with discrete and well-defined actions, such as selecting functions, actuating the gripper, or executing predefined sequences. However, speech recognition may be affected by environmental noise, microphone quality, pronunciation, accent, and inter-user variability [8]. Therefore, voice is well suited for activating well-defined functions but does not provide the same level of precision required for continuous control of a robotic manipulator's trajectory [9].

Multimodal interaction seeks to exploit the complementary characteristics of multiple communication modalities, preventing a single interface from handling all control functions. From this perspective, eye gaze can provide spatial or directional information, while voice can be used to execute specific actions, switch operating modes, or select predefined routines. In this way, continuous control is separated from discrete decision-making, the required gaze fixation time is reduced, and the user's visual attention remains focused on the robot during task execution [10].

In the field of assistive technology, several studies have also reported the use of hands-free interfaces to control assistive robotic manipulators. For example, Cio et al. employed eye-gaze tracking and stereo vision to perform reaching and manipulation tasks [11]. Similarly, Sunny et al. demonstrated that users could control a robotic manipulator mounted on a wheelchair through a gaze-based visual interface while performing activities of daily living [12]. Lin et al. investigated direct object selection using eye gaze to facilitate manipulation tasks for individuals with high-level spinal cord injuries [13]. Other studies evaluated multimodal hands-free control systems that combined different input signals to operate a robotic arm [14]. Likewise, Salinas-

Martínez et al. demonstrated that the combination of gestures and voice commands enables interaction with robotic systems [15]. Collectively, these studies confirm the feasibility of using eye gaze, voice, and other natural interaction modalities. However, they mainly focus on target selection, shared autonomy, virtual interfaces, and the general integration of multiple input signals.

Despite these advances, the assignment of a specific function to each modality during the execution of physical manipulation tasks has been scarcely investigated. An alternative approach consists of using eye gaze to generate directional commands and reserving voice commands for gripper actuation and the activation of automatic routines. This distribution avoids requiring eye gaze to be simultaneously used for observing the environment, controlling the manipulator, and confirming actions. Furthermore, it presents multimodality as a functional distribution of commands rather than as a redundant combination of signals.

In this context, this work presents a task-oriented multimodal control architecture for a five-degree-of-freedom assistive robotic manipulator equipped with a gripper actuated by a servomotor. Eye-gaze tracking is implemented in Python using MediaPipe Face Mesh, which enables the acquisition of horizontal and vertical gaze position information. These data are transmitted to MATLAB through UDP communication. In this environment, voice commands are recognized, the kinematic model is processed, joint limits are applied, and the manipulator trajectories are generated. Subsequently, joint positions are transmitted through serial communication to an ESP32, which is responsible for controlling the servomotors using PWM signals. The kinematic chain is composed of five joints. Additionally, an independent servomotor is used to open and close the gripper.

The main contribution of this work consists of a functional assignment strategy: eye-gaze tracking enables control of the movement direction, while voice commands control the gripper and activate automatic routines and pick-and-place operations. The proposed approach integrates perception, command recognition, kinematic coordination, trajectory generation, and physical actuation modules. These modules are organized into a distributed architecture implemented using Python, MATLAB, and ESP32. The operation of the proposed system is evaluated through eye-gaze classification tests, voice command recognition, inter-module communication, joint movements, gripper control, and task execution under controlled conditions.

II. Proposed Multimodal System

A. General system architecture

The system is organized within a three-level distributed architecture: the perceptual level, the processing level, and the action level. At the perceptual level, a camera captures the user's facial image, and this image is processed by Python using MediaPipe Face Mesh to calculate the variables D_X and D_Y which are associated with the gaze direction. This information is transmitted to MATLAB through the UDP protocol. MATLAB is responsible for integrating the gaze signals with the voice commands recognized using Wav2Vec2; simultaneously, MATLAB manages the manipulator kinematic model, applies joint limits, and generates the corresponding trajectories.

The obtained angles are transmitted through serial communication to the ESP32, which generates the PWM signals to actuate the five joints of the manipulator, as well as the end-effector gripper. In this way, eye gaze enables the control of the manipulator's directional movements, while voice commands control the opening and closing of the gripper and activate the automatic manipulation routines.

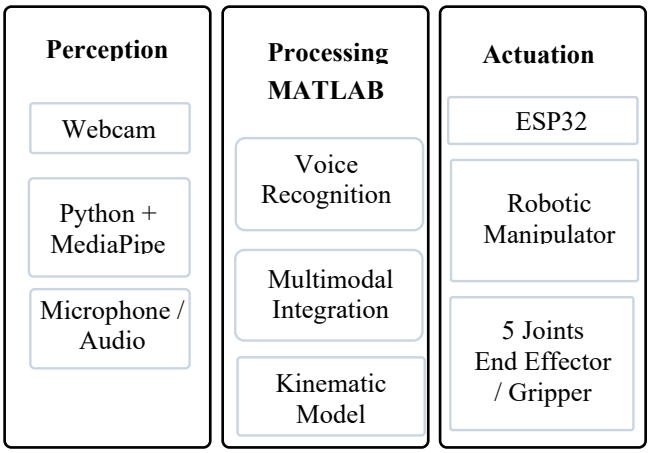


Fig. 1. General architecture of the multimodal control system.

B. Robotic manipulator configuration

The physical platform consists of a five-degree-of-freedom serial manipulator, composed of revolute joints and equipped with an actuated gripper as an end-effector tool. The main joints are driven by MG996R servomotors, while the wrist and gripper movements are achieved using MG90S microservomotors. In summary, the system is based on six actuators: five responsible for defining the manipulator configuration, plus one additional actuator that enables the opening and closing of the gripper.

Taking as reference the Denavit–Hartenberg parameters presented in Table I, where the values of a_i and d_i (in meters) are $d_1 = 0.14$ m, $a_2 = 0.13$ m, $a_3 = 0.10$ m, $d_4 = 0.03$ m, Tool $(d_{tool}) = 0.13$ m.

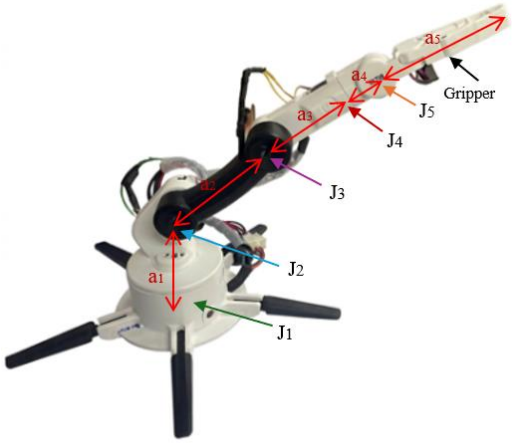


Fig. 2. Kinematic configuration of the 5-DOF assistive robotic manipulator with an actuated gripper.

TABLE I. DENAVIT–HARTENBERG PARAMETERS OF THE MANIPULATOR

i	a_i(m)	α_i(rad)	d_i(m)	θ_i
1	0	$\pi/2$	0.14	θ_1
2	0.13	0	0	θ_2
3	0.10	0	0	θ_3
4	0	$\pi/2$	0.03	θ_4
5	0	$-\pi/2$	0	θ_5
Tool	0	0	0.13	0

C. Eye-Gaze Tracking and Speech Recognition

The eye-gaze tracking module was implemented in Python using MediaPipe Face Mesh. Based on the image captured by the camera, the system detects the face and locates the reference points corresponding to the eyes. With this information, the variables D_X and D_Y are calculated and used to classify the gaze into five states: left, right, up, down, and center. To prevent small variations or natural movements from generating incorrect commands, a central tolerance zone was established. When the gaze remains within this zone, the system maintains the current manipulator position. If the face or eye landmarks are not detected correctly, no movement command is generated.

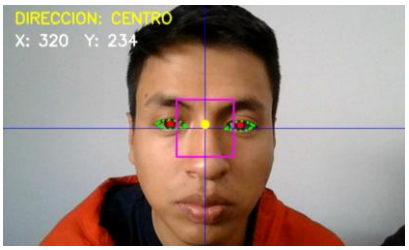


Fig. 3. Facial landmark tracking and generation of the D_X and D_Y variables for gaze classification.

Speech recognition is executed in MATLAB using Wav2Vec2 trained in Spanish. The vocabulary is limited to the commands ABRIR, CERRAR, UNO, DOS, and TRES. The first two commands actuate the gripper, while the remaining three activate previously defined automatic routines. Words not included in the authorized vocabulary are discarded and do not generate movement. In this way, eye gaze is used for the directional control of the manipulator, while voice is used for discrete actions and task selection.

D. Task-Oriented Multimodal Strategy

The proposed interaction strategy is as follows: each interaction modality is associated with a specific function within the manipulation task. Thus, eye-gaze tracking is intended to issue directional commands for the manipulator, while voice is intended to issue commands for discrete actions. The left, right, up, and down states progressively modify the joint positions, whereas the center state maintains the joint configuration unchanged.

The ABRIR and CERRAR commands control the end-effector, while the UNO, DOS, and TRES commands enable previously stored automatic trajectories. Before executing a command, MATLAB verifies the joint limits and validates the received state. The resulting positions are sent to the ESP32 for execution through PWM signals.

With this distribution, eye gaze is not simultaneously used to observe the environment, move the robot, and confirm actions. Therefore, multimodality is formulated as a functional task assignment rather than as a redundant combination of signals. Eye gaze is dedicated to the directional control of the manipulator, while voice controls the gripper and activates the automatic pick-and-place routines.

III. EXPERIMENTAL EVALUATION

The evaluation was conducted under controlled conditions to verify the individual and integrated operation of the eye-gaze tracking, speech recognition, communication, and robotic actuation modules. The tests focused on determining whether the commands generated by the user were correctly interpreted and executed by the manipulator during basic pick-and-place tasks.

A. Experimental Setup

The experimental system consisted of a webcam, a computer-integrated microphone, processing modules implemented in Python and MATLAB, an ESP32 controller, and the five-degree-of-freedom manipulator with an actuated gripper. The camera was positioned in front of the user to capture facial features and eye movements, while the microphone was used to receive voice commands.

The tests were conducted in an indoor environment with controlled lighting conditions and with the user positioned in front of the camera. Lightweight objects were placed within the robot's workspace at reachable positions. Communication between Python and MATLAB was performed through UDP, while joint positions were transmitted from MATLAB to ESP32 through serial communication. The experimental setup used during the tests is presented in Fig. 4.



Fig. 4. Experimental setup during the interaction of a participant with the robotic manipulator through eye-gaze tracking and voice commands.

B. Evaluation Protocol

The protocol was divided into four stages. First, gaze classification was verified for the left, right, up, down, and center states. Second, the recognition of the ABRIR, CERRAR, UNO, DOS, and TRES commands was evaluated, verifying that each recognized word generated the programmed action.

Subsequently, the transmission of the D_X and D_Y variables through UDP and the transmission of joint positions through serial communication were verified. Finally, the integrated operation was evaluated through directional control of the manipulator, gripper opening and closing, and the execution of

three automatic routines. The pick-and-place sequences comprised the stages of approach, grasping, transportation, release, and return to a safe position, as shown in Fig. 6.



Fig. 5. Functional tests of the multimodal system with different participants.

A ten-question survey was administered to twenty participants without motor disabilities to assess system usability, responsiveness, and task execution. Each participant completed a three-stage evaluation procedure. First, the operating principles of the multimodal interface were explained. Second, participants performed a practice trial to become familiar with the eye-gaze and voice commands. Finally, an evaluation trial was conducted to assess their ability to control the manipulator and execute the programmed tasks.

Since the participants did not belong to the intended target population, the results represent a preliminary evaluation under controlled conditions rather than a clinical validation. Individuals with motor impairments may exhibit different gaze-control capabilities, head movements, and speech characteristics. Therefore, the reported results should not be generalized to users with motor disabilities. Future work will include evaluations with the target population under more representative conditions.

C. *Evaluation criteria*

The operation was considered correct when: 1) the eye gaze direction was classified according to the requested orientation; 2) the voice command was associated with the corresponding action; 3) the data were correctly transmitted between the modules; 4) the joints and the gripper responded to the received commands; and 5) the automatic routine completed the programmed sequence without interruptions.

The results were recorded through functional observation and organized according to the compliance of each module. A five-level Likert scale was used for the survey, and the overall mean and the proportion of ratings within the highest categories were calculated.

The reported recognition accuracies and routine success rates correspond to preliminary functional evaluations conducted under controlled experimental conditions. Although twenty participants completed ten multimodal interaction trials each, the detailed trial-level records used to calculate the individual recognition and success metrics were not retained. Consequently, the reported percentages should be interpreted as descriptive results of the preliminary evaluation rather than independently reproducible performance estimates. Future evaluations will incorporate systematic trial logging, confusion matrices, and explicit reporting of successful and unsuccessful attempts for each interaction modality.

The Cartesian positioning performance was experimentally evaluated by executing each predefined trajectory approximately five times. During these trials, a graduated ruler was used to measure the deviation between the desired and actual end-effector positions. The recorded measurements were used to assess positioning accuracy and consistency across repeated executions. Since the manipulator operates without external position feedback sensors, these measurements provide an approximate assessment of positioning performance under controlled experimental conditions.

IV. Results And Discussion

A. *Eye-Gaze Tracking and Speech Recognition*

The eye-gaze tracking module detected the face and the eye landmarks required to calculate the D_X and D_Y variables, which were used to classify the gaze into left, right, up, down, and center directions. The overall accuracy reached 96.4%, with the best performance obtained for the center direction (98.1%). The main confusions occurred between the vertical directions due to the smaller separation between upward and downward eye movements.

Speech recognition achieved an overall accuracy of 98.0% for the ABRIR, CERRAR, UNO, DOS, and TRES commands. The errors were mainly associated with pronunciation variations and pauses between consecutive commands. These results support the complementary assignment of modalities: eye gaze for directional control and voice for discrete actions.

B. *Communication and manipulator response*

UDP communication enabled the transmission of the D_X and D_Y variables from Python to MATLAB, while the serial link transmitted the joint positions to the ESP32. The measured total latency was 102ms, which allowed an adequate response for manipulator operation under controlled conditions.

The mean Cartesian error was 2.6mm, with a standard deviation of 0.8mm. The observed deviations were mainly associated with the mechanical tolerances of the servomotors and the absence of position feedback sensors.

C. *Automatic routine execution*

The UNO, DOS, and TRES routines completed the approach, grasping, transportation, release, and return stages. The success rate was 98.3%, with an average execution time of 9.4s.

The repetition of the same trajectory resulted in an RMSE of 2.3 mm, indicating stable behavior during the tests. Overall, these

results show that the multimodal strategy enables the combination of directional adjustments through eye gaze with the activation of automatic tasks through voice commands.

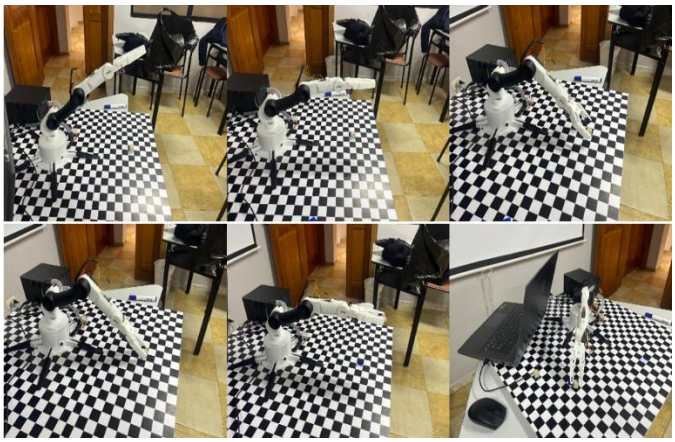


Fig. 6. Physical execution of the programmed automatic trajectories.

D. Participant Evaluation

The survey applied to twenty participants consisted of ten questions and generated 200 ratings. A total of 81.5% of the responses were concentrated in categories 4 and 5 of the Likert scale, while the overall mean was 3.90/5.

These results reflect a favorable perception of the system operation. However, due to the limited number of participants and the tests being conducted under controlled conditions, the evaluation should be considered a preliminary technical validation.

TABLE II. SUMMARY OF THE FUNCTIONAL RESULTS OF THE SYSTEM.

Metric	Result
Overall accuracy of eye-gaze tracking	96.40%
Overall accuracy of speech recognition	98.00%
Total latency	102 ms
Mean Cartesian error	2.6 mm
Standard deviation	0.8 mm
Routine success rate	98.30%
Average execution time	9.4 s
Repeatability RMSE	2.3 mm
Survey mean	3.90/5
Ratings in categories 4 and 5	81.50%

An important limitation of the current prototype is the use of low-cost RC servomotors, which exhibit mechanical backlash and lack direct torque feedback. Although joint limits are verified in MATLAB before executing movement commands, the system does not incorporate an emergency stop mechanism, active force limitation, or collision detection. Consequently, the prototype is intended exclusively for supervised experimental evaluation under controlled conditions and is not suitable for unsupervised assistive applications. Future improvements will include an emergency stop system, force and position sensing, collision detection, and additional mechanical safety measures to ensure safer physical interaction with users.

The distributed architecture based on Python, MATLAB, and ESP32 introduces potential communication limitations associated with UDP packet loss, serial transmission delays, and software synchronization. In the current implementation, the manipulator completes the last received movement command and maintains its final joint configuration when no additional commands are received. However, this behavior does not constitute a dedicated communication fault-detection mechanism. Future improvements will consider communication watchdogs, command acknowledgment protocols, and the integration of perception and control modules into a unified edge-computing platform to improve system reliability and reduce communication dependencies.

Although the proposed multimodal architecture demonstrated functional feasibility under controlled conditions, the experimental evaluation was limited to the combined use of eye-gaze tracking and voice commands. No direct comparison with eye-gaze-only or voice-only control interfaces was performed. Therefore, the current results do not establish a quantitative advantage over unimodal approaches. Future work will include comparative experiments to evaluate task completion time, recognition accuracy, and user workload across different interaction modalities.

In general terms, the results support the functional feasibility of the task-oriented multimodal strategy. The differentiated assignment of commands prevents a single modality from concentrating the entire operation; however, illumination, noise, mechanical accuracy, and the absence of sensory feedback remain the main limitations of the prototype.

V. CONCLUSIONS

A task-oriented multimodal control strategy was implemented for a five-degree-of-freedom assistive robotic manipulator with an actuated gripper. The distributed architecture based on Python, MATLAB, and ESP32 enabled the integration of eye-gaze tracking, speech recognition, kinematic coordination, trajectory generation, and physical actuation.

The functional assignment of modalities enabled the use of eye gaze for directional control and voice for gripper actuation and the activation of automatic routines. The system achieved an overall accuracy of 96.4% in gaze classification and 98.0% in voice command recognition, with a total latency of 102ms.

In the automatic tasks, a success rate of 98.3%, an average execution time of 9.4 s, and a repeatability RMSE of 2.3 mm were achieved. Likewise, the mean Cartesian error was 2.6 mm, demonstrating stable behavior for basic lightweight object manipulation tasks under controlled conditions.

The evaluation with twenty participants produced an overall mean of 3.90/5, with 81.5% of the responses in categories 4 and 5. These results support the technical feasibility of the proposed strategy; however, the validation remains preliminary due to the limited number of participants, the absence of sensory feedback, and the tests being conducted only in a controlled environment. As future work, the incorporation of position and force sensors, the expansion of the experimental evaluation, and the execution of tests with users belonging to the target population are proposed.

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